EX NO : 5 29/09/23	DESIGN AND IMPLEMENTATION OF PARITY GENERATOR (CHECKERS)

AIM

To design and implement 3-bit parity generators and checkers.

APPARATUS REQUIRED

S.NO	NAME OF THE COMPONENT	SPECIFICATION	QUANTITY
1.	IC trainer kit	-	1
2.	Logic gates IC's	IC 7486	1
3.	Connecting wire	-	1 set.

THEORY

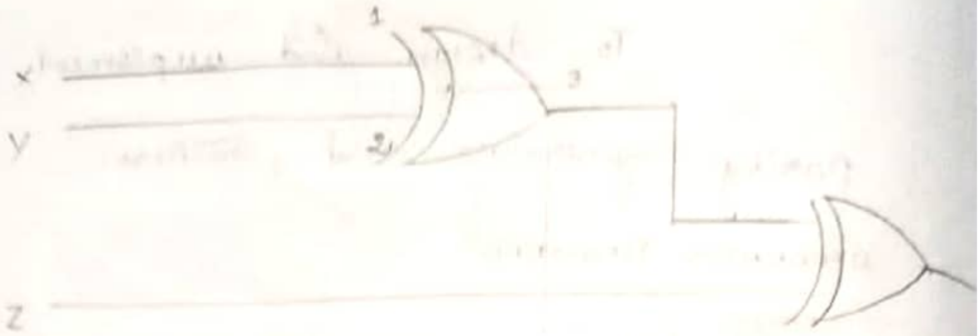
* A parity bit is used for the purpose of detecting error during transmission of binary information.

* A parity bit is an extra single bit included with a message.

* An error is detected if the checker parity does not correspond with the one transmitted.

* The parity which is transmitted along with the message is called parity.

Even Parity generator logic diagram



$$P = X \oplus Y (Z \oplus P)$$

TRUTH TABLE

X	Y	Z	P
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

generator the parity which is generated received is called parity checker.

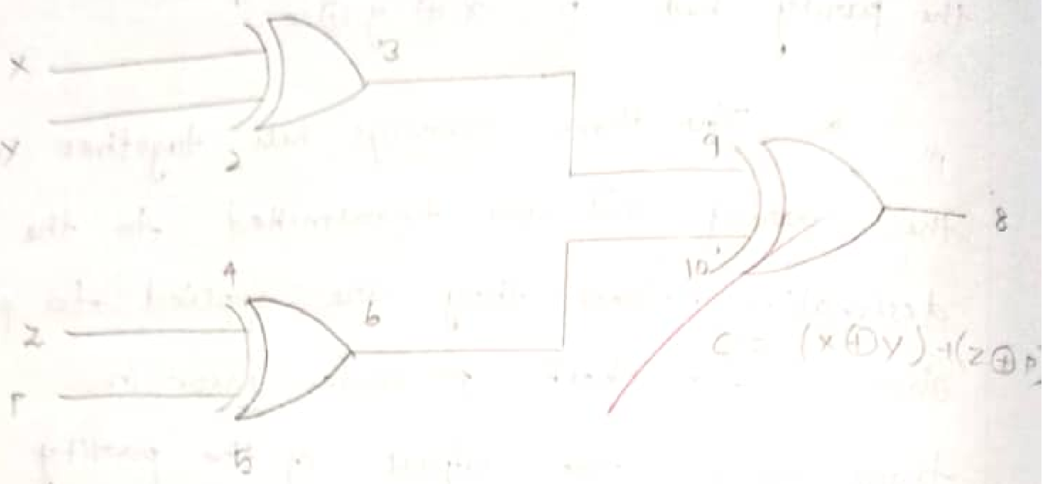
the parity bit p, $x \oplus y \oplus z$.

* The three message bits together which the parity bit are transmitted to the destination where they are applied to parity checker to check possible error in transmission the output of the parity checker should have even number of ones for accurate transmission.

* If the error is encounter then it will have odd number of ones the parity checker can be implemented with exclusive or OR gates.

$$c = x \oplus y (z \oplus p)$$

EVEN PARITY CHECKER LOGIC DIAGRAM.



TRUTH TABLE

X	Y	Z	P	C
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

verified

PROCEDURE

* Connections are made as per the circuit diagram.
* Switch ON IC trainer kit.
* Find the logic 0 or 1 to the input variable.
* The output is verified using the LED indicators.

RESULT

29/9/23

Thus 3 bit parity generator and checkers were designers and implemented.

Ex No : 6 29/09/23	DESIGN AND IMPLEMENTATION OF PRIORITY ENCODERS USING LOGIC GATES.

AIM

To design and implement priority encoder using logic gates

APPARATUS REQUIRED

S.NO	NAME OF THE COMPONENTS	SPECIFICATION	QUANTITY
1.	IC trainer kit	-	1
2.	Logic gates IC's	IC 7404 IC 7432	1
3.	Logic gates IC's	IC 7408	2
4.	Connecting wires	-	1 set

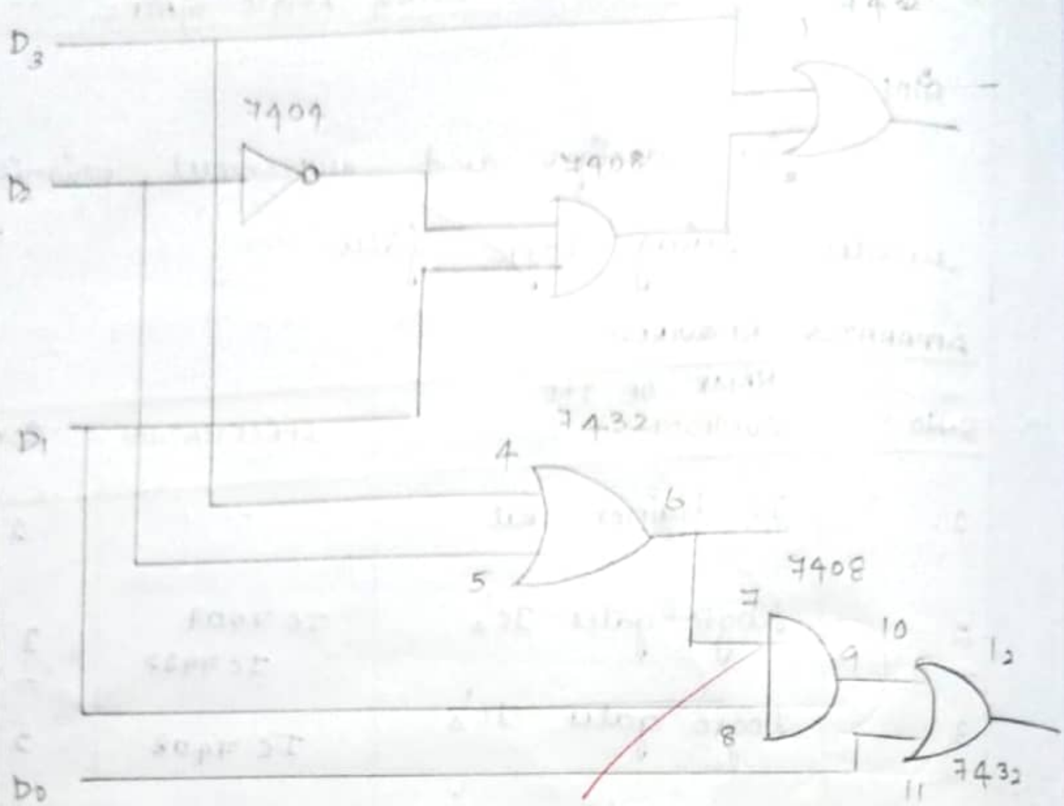
THEORY

* An encoder is digital circuit that performs the inverse operations of the decoder.

* It has 2n input and n output that represent the code of the order of the input that is set to one.

* this encoder has two main draw backs. when more than one input is not same time then output could indicate a wrong code.

LOGIC DIAGRAM



TRUTH TABLE

INPUT				OUTPUT		
D_0	D_1	D_2	D_3	X	Y	Z
0	0	0	0	X	X	0
1	0	0	0	0	0	1
X	1	0	0	0	1	1
X	X	1	0	1	0	1
X	X	X	1	1	1	1

eg. D5 + D6 are one at the same terms there and are indicating D4 is one to If no input is one which is not valid code, there are all zero which indicate a code do to overcome three priority encoder.

PROCEDURE

* Connections are made as per the circuit diagram.

* Switch ON the IC trainer kit.

* Apply the logic input 0 or 1 to the binary of priority encoder.

* Verify the truth table by observing the output indicators.

RESULT

24/9/23

Thus the 4 to 2 priority encoder was designed and implemented using logic gates and their operations are verified.